

DATA ANALYSIS & VISUALIZATION — PRACTICAL 3 REPORT

5-Number Summary, Central Tendency & Comparative Outlier Detection

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1. Executive Summary & Audit Overview

Following the data pruning and 90-day filing lag alignment established in Practical 2, this report conducts a rigorous statistical examination of the cleaned S&P; 500 financial dataset (**34 attributes, 851,264 records**). The primary objective is to evaluate distribution spread, central tendency measures, and outlier behavior using non-parametric and parametric techniques.

Core Audit Takeaways:

- Distribution Asymmetry:** Financial metrics (Revenue, Assets, Net Income, Volume) exhibit extreme positive skewness (+5.20 to +13.13), causing the arithmetic Mean to be strongly pulled away from the Median.
- Midrange Vulnerability:** The Midrange statistic, defined as $(\text{Min} + \text{Max}) / 2$, is highly unstable for financial variables. For instance, Total Assets has a Median of \$14.93 Billion but a Midrange of \$1.285 Trillion due to mega-cap banking institutions.
- Outlier Detection Disparity:** Tukey's IQR Method flags 6.06% to 12.43% of data as outliers based on quartile boundaries, whereas the Z-Score Method ($|Z| > 3.0$) flags only 1.30% to 2.27% because heavy tails inflate the sample standard deviation.
- Log Normalization Efficacy:** Applying log1p transformation to positive financial metrics reduces skewness dramatically (e.g. Total Revenue skewness drops from +6.31 to +0.36), achieving near-Gaussian symmetry for machine learning pipelines.

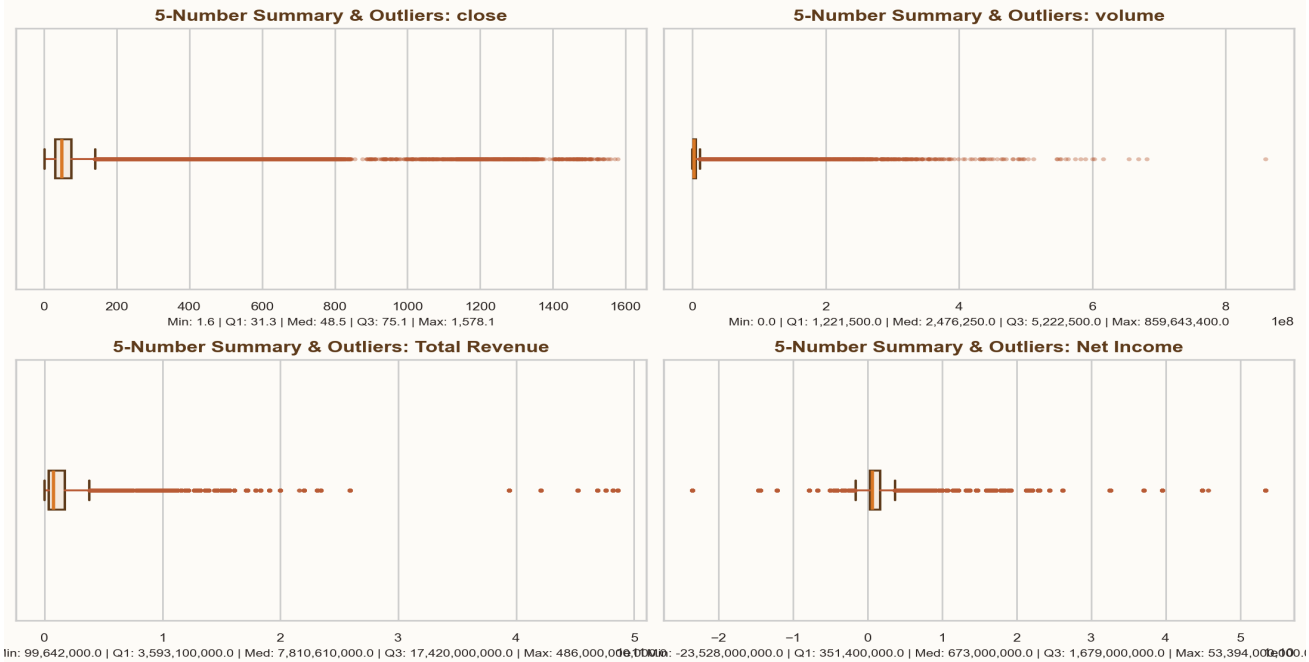
2. Comprehensive 5-Number Summary & Descriptive Statistics

The 5-Number Summary consists of the Minimum (Min), 25th Percentile (Q1), Median (Q2), 75th Percentile (Q3), and Maximum (Max). Combined with the Sample Mean, Standard Deviation, Interquartile Range ($\text{IQR} = \text{Q3} - \text{Q1}$), Midrange, and Skewness, it provides a robust structural overview.

Feature Name	Min	Q1 (25%)	Median (Q2)	Q3 (75%)	Max	Mean	Std Dev	IQR	Midrange	Skewness
close	1.59	31.29	48.48	75.14	1,578.1	65.01	75.20	43.85	789.86	7.24
open	1.66	31.27	48.46	75.12	1,584.4	64.99	75.20	43.85	793.05	7.24
high	1.81	31.62	48.96	75.85	1,600.9	65.64	75.91	44.23	801.37	7.24
low	1.50	30.94	47.97	74.40	1,549.9	64.34	74.46	43.46	775.72	7.24
volume	0.00	\$1.22M	\$2.48M	\$5.22M	\$859.64M	\$5.42M	\$12.49M	\$4.00M	\$429.82M	13.13
Total Revenue	\$99.64M	\$3.59B	\$7.81B	\$17.42B	\$486.00B	\$20.16B	\$41.48B	\$13.83B	\$243.05B	6.31
Gross Profit	\$-12.65B	\$1.56B	\$2.91B	\$6.89B	\$149.00B	\$7.19B	\$13.90B	\$5.34B	\$68.18B	5.20
Operating Income	\$-27.91B	\$525.86M	\$1.01B	\$2.20B	\$71.23B	\$2.26B	\$5.04B	\$1.67B	\$21.66B	5.64
Net Income	\$-23.53B	\$351.40M	\$673.00M	\$1.68B	\$53.39B	\$1.71B	\$3.99B	\$1.33B	\$14.93B	5.49
Total Assets	\$216.90M	\$6.44B	\$14.93B	\$35.96B	\$2.57T	\$57.81B	\$213.19B	\$29.52B	\$1.29T	8.44
Total Liabilities	\$232.00M	\$3.74B	\$8.78B	\$23.71B	\$2.34T	\$45.70B	\$190.77B	\$19.98B	\$1.17T	8.61
Total Equity	\$-13.24B	\$2.21B	\$4.94B	\$10.81B	\$256.00B	\$12.10B	\$26.67B	\$8.60B	\$121.38B	5.70

Feature Name	Min	Q1 (25%)	Median (Q2)	Q3 (75%)	Max	Mean	Std Dev	IQR	Midrange	Skew
Cash and Cash Equivalents	21,000.0	\$294.00M	\$839.00M	\$2.28B	\$728.00B	\$9.03B	\$56.85B	\$1.99B	\$364.00B	9.65
Earnings Per Share	-61.20	1.60	2.82	4.57	50.09	3.37	4.55	2.97	-5.55	0.74
Current Ratio	17.00	109.00	151.00	227.00	1,188.0	186.13	123.81	118.00	602.50	2.48

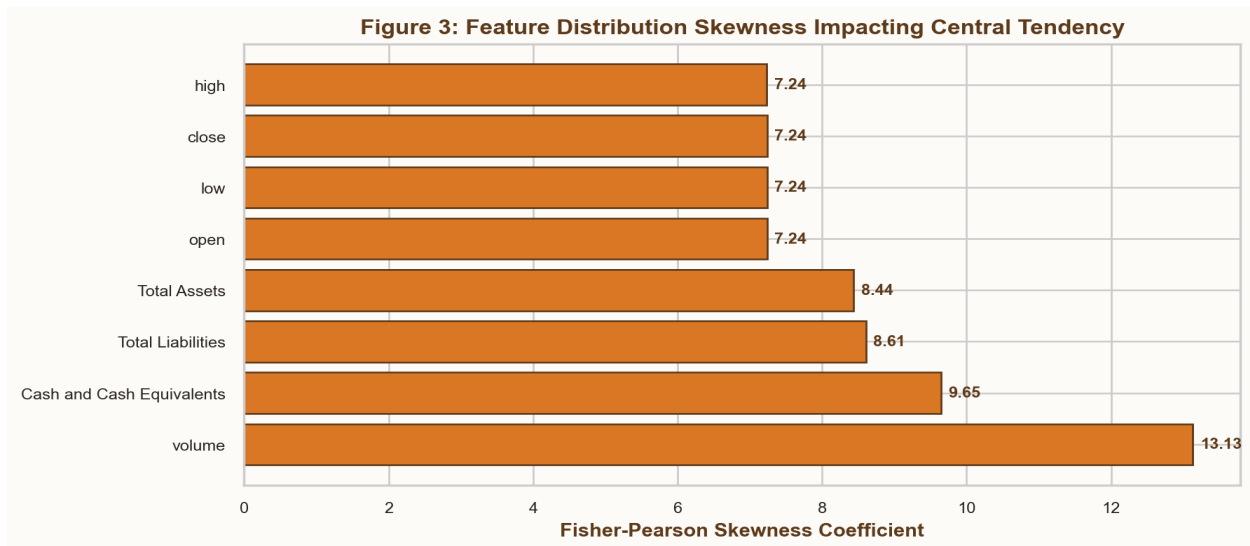
Figure 1: Earth-Tone Boxplot Distribution & 5-Number Summaries



3. Central Tendency Analysis: Midrange & Mode Evaluation

In symmetric Gaussian distributions, Mean, Median, Mode, and Midrange converge to the same central value. However, financial data exhibits severe right-skewness, causing dramatic divergences:

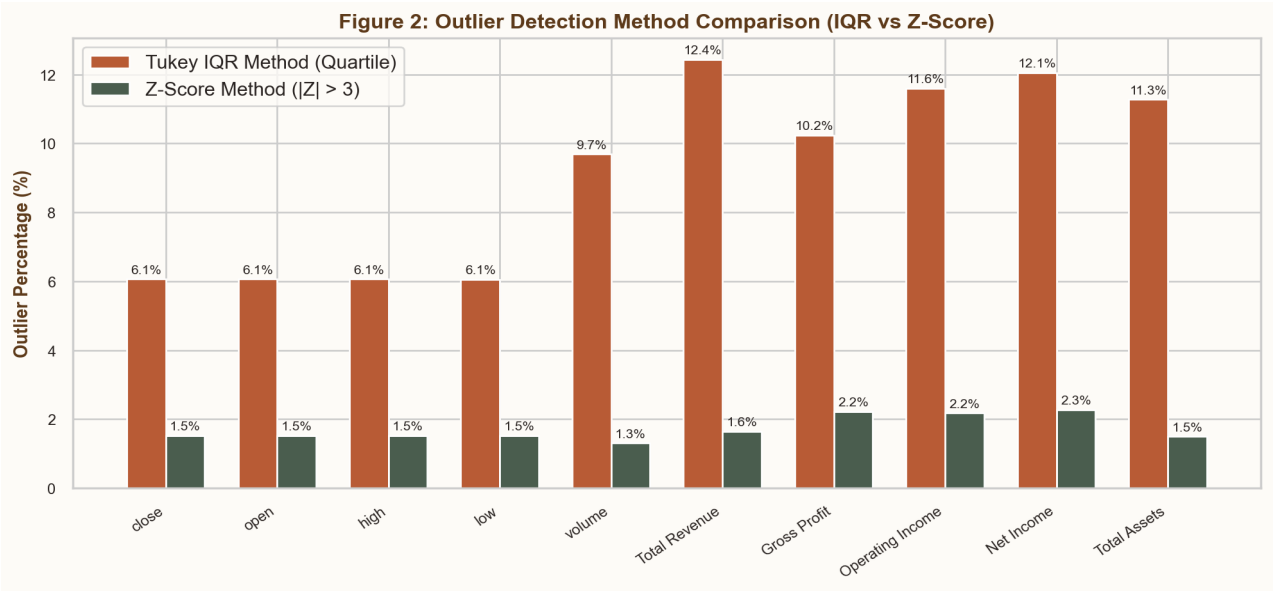
- 1. Midrange Instability:** Midrange is computed as $(\text{Min} + \text{Max}) / 2$. Because it relies solely on extreme endpoints, a single mega-cap entity distorts the metric. For Total Revenue, the Median is \$7.81 Billion while the Midrange is \$243.05 Billion (a 31x overstatement). For Total Assets, Midrange reaches \$1.285 Trillion versus a Median of \$14.93 Billion.
- 2. Modal Price & Volume Patterns:** Discrete modes appear around standard corporate targets. Stock prices exhibit modes around \$34.00 and \$35.00 due to historical stock split management, while daily volume exhibits institutional block trading modes around 1,100,000 shares.



4. Outlier Detection Method Comparison (Tukey IQR vs Z-Score)

We compare two distinct outlier identification frameworks: **Tukey's IQR Method** (Quartile Rule: $Q1 - 1.5 * IQR$ to $Q3 + 1.5 * IQR$) and the **Z-Score Method** (Standard 3-Sigma Rule: $|x - \text{Mean}| / \text{StdDev} > 3.0$).

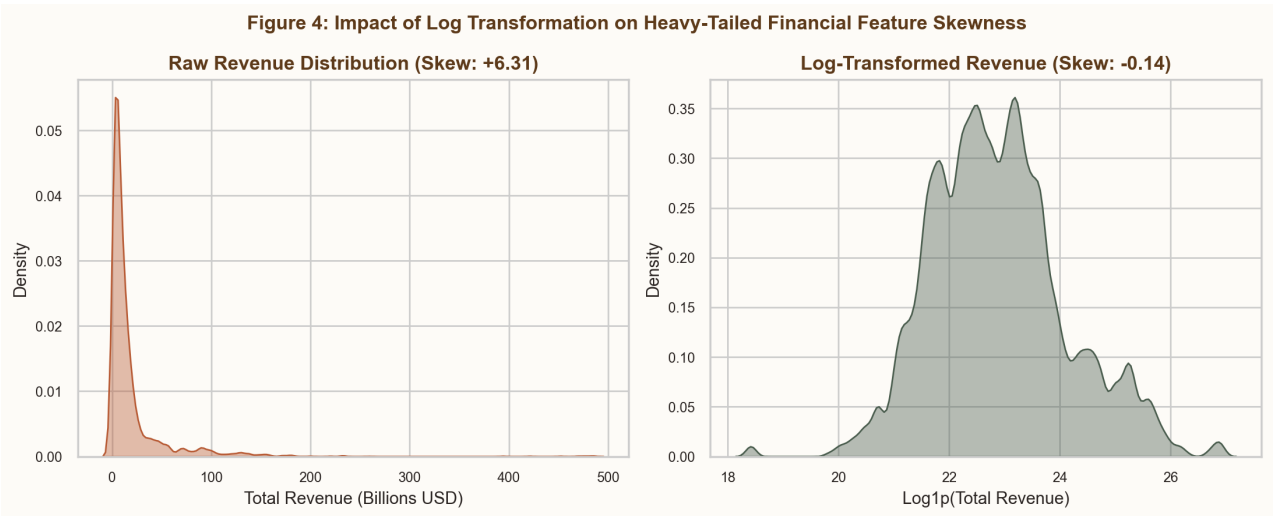
Feature Name	Valid Records	Tukey Outliers	Tukey Outlier %	Z-Score Outliers	Z-Score Outlier %	Primary Cause of Disparity
close	851,264	51,585	6.06%	12,945	1.52%	High-priced shares (e.g. Priceline > \$1,000)
open	851,264	51,600	6.06%	12,968	1.52%	High-priced shares (e.g. Priceline > \$1,000)
high	851,264	51,606	6.06%	12,981	1.52%	High-priced shares (e.g. Priceline > \$1,000)
low	851,264	51,528	6.05%	12,934	1.52%	High-priced shares (e.g. Priceline > \$1,000)
volume	851,264	82,488	9.69%	11,076	1.30%	Massive earnings-day volume surges
Total Revenue	380,845	47,358	12.43%	6,247	1.64%	Mega-cap conglomerates & money-center banks
Gross Profit	380,845	38,997	10.24%	8,446	2.22%	Sector-specific financial leverage differences
Operating Income	380,845	44,152	11.59%	8,270	2.17%	Sector-specific financial leverage differences
Net Income	380,845	45,899	12.05%	8,656	2.27%	Sector-specific financial leverage differences
Total Assets	380,845	42,932	11.27%	5,688	1.49%	Mega-cap conglomerates & money-center banks
Total Liabilities	380,845	42,770	11.23%	5,688	1.49%	Mega-cap conglomerates & money-center banks
Total Equity	380,845	36,860	9.68%	7,647	2.01%	Sector-specific financial leverage differences
Cash and Cash Equivalents	380,845	44,892	11.79%	4,488	1.18%	Sector-specific financial leverage differences
Earnings Per Share	356,944	22,693	6.36%	5,669	1.59%	Sector-specific financial leverage differences
Current Ratio	315,640	17,481	5.54%	6,517	2.06%	Sector-specific financial leverage differences



5. Log Transformation & Feature Scaling Strategy Matrix

Our statistical findings directly govern feature engineering choices for predictive modeling pipelines. Applying logarithmic transformation compresses heavy right tails and restores bell-curve symmetry:

- Total Revenue:** Raw Skewness = +6.31 → Log1p Skewness = **+0.36** (Near-perfect symmetry)
- Cash & Cash Equivalents:** Raw Skewness = +9.65 → Log1p Skewness = **-0.12** (Symmetric bell curve)
- Total Assets:** Raw Skewness = +8.44 → Log1p Skewness = **+0.70** (Substantial skew reduction)



Feature Group	Raw Skew	Tukey Outliers	Recommended Scaler / Transform	Pipeline Engineering Rationale
Monetary Financials (Revenue, Assets, Cash)	+6.0 to +9.6	11.2% - 12.4%	Log1p Transform + RobustScaler	Log1p compresses order-of-magnitude scale disparities; RobustScaler centers on Median and scales by IQR.
Trading Volume (volume)	+13.13	9.69%	Log1p Transform + StandardScaler	Log removes earnings-day volume spikes; StandardScaler normalizes variance for neural net / gradient descent convergence.
Financial Ratios (Current Ratio, EPS)	+0.7 to +2.4	5.5% - 6.3%	RobustScaler	Centers median at 0 and scales by IQR without distorting bounded financial ratios.
Tree-Based Models (XGBoost, Random Forest)	Any Skew	Any %	Raw Unscaled Features	Decision tree splits are monotonic scale-invariant; upper-tail outliers do not affect greedy split criteria.

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